

JIHEON KANG

PROFILE

Electrical and Electronic Engineering student at Yonsei University working on medical AI, particularly genomics and AI-driven drug discovery. I started asking biological questions to move beyond artificial tasks toward real ones: finding the rules nature writes into DNA, cells, and proteins is what excites me.

CONTACT INFORMATION

- **Email:** danieljiheon@yonsei.ac.kr
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RESEARCH INTERESTS

- Genomics AI
- Multimodal AI
- Model Interpretability

EDUCATION

B.S. in Electrical and Electronic Engineering
Yonsei University

Expected Feb. 2028

PUBLICATIONS & RESEARCH

Layer-wise Settling Depth Reveals Biological Grammar in Genomic Foundation Models
Accepted for Oral Presentation, ICML GenBio Workshop — bioRxiv link

2026

- Yoonjin Cho, **Jiheon Kang**, Subin Park, and Prof. Sangwoo Kim.
- Training-free analysis of the layer where Evo 2 settles on each nucleotide; splice sites settle earlier than introns.
- Follow-up research on Evo 2 — *ongoing*.

Maximizing Frame-Level Video Understanding for Efficient Video Temporal Grounding
Accepted, 2025 IEIE Summer Annual Conference

2025

- **Jiheon Kang**, Suyong Kim, Harim Noh, and Heejae Yang.
- Training-free video temporal grounding with one frozen image model (CLIP).

Bi-CoT: Forward Reasoning and Reverse Verification for Explainable Multi-Hop QA
Accepted, The 6th Korea Artificial Intelligence Conference

2025

- **Jiheon Kang** and Suhwan Jung.
- Inference-only multi-hop QA with step-wise evidence checks and reverse verification, improving answer accuracy.

MEDICAL & GENOMICS PROJECTS

CAFA 6 Protein Function Prediction <i>Bronze Medal, International Protein Function Prediction Challenge</i> - Predicted protein function (Gene Ontology terms) from sequence; worked on the ProtT5 protein language model branch. - Jiheon Kang, Minseon Koo, Yoonjin Cho (Team Lead), Subin Park, and Yumin Jung.	2026
Virtual Cell Challenge 2026 <i>Ongoing, Zero-Shot Perturbation Prediction Project</i> - Built a training-free pipeline that transfers gene-knockdown effects from similar cell lines to unseen ones. - Jiheon Kang (Team Lead), Minseok Kim, Yumin Jung, Seojin Kim, Donghyun Lee, and Seoyeon Kim.	2026–Ongoing
PhenoFocus: AI-Driven Drug Discovery from Cell Images <i>Advanced to the Main Round, 2026 Research-Idea Commercialization Challenge</i> - Built a scaffold-hopping MVP that retrieved a same-mechanism HDAC inhibitor from a different chemical class. - Jiheon Kang (Team Lead), Jaeyoung Ha, Jungyoon Han, and Hyunjun Hwang.	2026–Ongoing
GeoFlowAgent: A Tool-Using AI Agent for Genomic Variant Interpretation <i>Independent Research Project</i> - Built an AI agent that uses frozen biomedical models to plan multi-step genomics tool use for variant interpretation. - Showed that replanning after observing each tool result clearly outperforms one-shot planning. - Jiheon Kang (independent project).	2026
Lightweight BU-Net Implementation <i>Medical AI Model Optimization — presented at the deep daiv. Open Seminar</i> - Built a lightweight BU-Net for limited GPUs and compared it with U-Net on brain-tumor MRI segmentation. - Jiheon Kang (Team Lead), Boyoung Kwon, and Jaeryeong Hwang.	2024

LEADERSHIP & ACTIVITIES

Yonsei AI (YAI) <i>Vice President & Functional Genomics Team Lead of Yonsei University AI Academy</i> - YAI Member (January 2026 – Present). - YAI Vice President (July 2026 – Present). - Functional Genomics Team Lead (June 2026–Present).	2026–Present
deep daiv. <i>Member — Led team projects across diverse AI domains</i> - Video Temporal Grounding research (Video-Language Model, Temporal Localization) — Team Lead; led to an IEIE paper. - RAG-grounded persona chatbot (retrieval over wiki pages, Prompt Engineering, LLMs) — Team Lead. - Local restaurant recommendation system development (Recommendation Systems, Data Preprocessing) — Team Lead.	2023–2025
deep daiv. Contents Team <i>AI Newsletter Writer & Magazine Editor</i> - Weekly AI Newsletter Writer (Mar. 2025–Jan. 2026): explained emerging multimodal AI papers with everyday analogies. - AI Magazine Editor (Aug.–Dec. 2024): edited and reviewed articles on AI and wrote pieces for beginners.	2024–2026